Observation Project: The Moon Diary and the Shadow Stick

Practical project alongside Lectures 1.1–1.6 4 weeks

Name: _____ Date: _____ Marks: _____ / 50

Why this project?

Everything in Unit 1 — the Moon's 50-minute daily delay, tithis, phases, local noon, the Sun's changing height — can be **discovered by yourself** with only your eyes, a stick and a notebook. This is how astronomy began.

Safety first

Never look directly at the Sun — in Part B you look only at the **shadow** on the ground. Observe the night sky with a friend or family member, and tell someone where you are.

Part A — The Moon Diary

30 marks

1. Start **2–3 days after Amavasya**: the young crescent appears low in the **west just after sunset**.
2. Observe the Moon on **at least 12 different days** over 4 weeks, at about the same clock time (e.g. 7:30 PM) for the first two weeks. After Purnima the Moon rises later, so look late at night or early in the morning (the waning Moon is visible in the morning sky — even in daylight!).
3. Measure **direction** with a compass app or landmarks, and **height** with your hand at arm's length (fist ≈ 10°, three fingers ≈ 5°, one finger ≈ 1°).
4. Look up the **tithi** in a panchang or calendar *after* you have made your sketch.

#	Date	Time	Direction	Height (°)	Sketch (shade the dark part)	% lit	Tithi
1							
2							
3							
4							
5							
6							
7							

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(continued)

#	Date	Time	Direction	Height (°)	Sketch (shade the dark part)	% lit	Tithi
8							
9							
10							
11							
12							
13							
14							

Questions to answer from YOUR data

1. **The daily shift** (4). On consecutive evenings at the same clock time, how far east did the Moon move in 24 hours? (Expected about 12°–13°.) What does this tell you about the Moon's orbit?
2. **The lit side** (3). On which side was the Moon lit while waxing? While waning? Where was the Sun each time?
3. **Phase vs tithi** (4). Plot tithi number (x-axis) against % lit (y-axis). Describe the shape. On which tithi was the Moon half-lit?
4. **Rising late** (3). If you saw the Moon rise on two different dates, by how much did the rising time change per day on average? Compare with ∼50 minutes.
5. **Earthshine** (1). Did the "dark" part of the crescent glow faintly? This is sunlight reflected from Earth onto the Moon. When was it most visible?

Marking: log (12) · sketches (5) · questions (13).

Part B — The Shadow Stick: local noon and true north **20 marks**

You need: a straight stick (∼1 m), flat sunny ground (11 AM–2 PM), a plumb line (a stone on a thread) to check the stick is vertical, pebbles or chalk, a measuring tape and a watch showing IST.

1. Fix the stick **vertically**; measure its height H above the ground.
2. From 11:30 AM to 1:30 PM, mark the tip of the shadow **every 10 minutes** and write the time beside each mark.
3. Measure each shadow length L from the base of the stick.
4. The **shortest shadow** gives your **local noon**; the line from the stick to that tip is the **north–south line**.

Time (IST)	Shadow L (cm)	Time (IST)	Shadow L (cm)
11:30		12:40	

(continued)

Time (IST)	Shadow L (cm)	Time (IST)	Shadow L (cm)
11:40		12:50	
11:50		13:00	
12:00		13:10	
12:10		13:20	
12:20		13:30	
12:30		Stick height H =	cm

Calculations

$$\text{Local noon (IST)} \approx 12{:}00 - 4\ \text{min} \times (\lambda - 82.5^\circ) \qquad \tan h = \frac{H}{L} \;\Rightarrow\; h = \tan^{-1}\frac{H}{L}$$

$$\varphi = 90^\circ - h + \delta \quad \text{(shadow pointing north)}$$

1. **Local noon** (3). At what IST was the shadow shortest? Compare with the prediction from your longitude; explain any difference (the equation of time can be up to $\pm$16 min).
2. **Sun's altitude** (4). Find h from the shortest shadow (calculator or table below).
3. **Your latitude** (4). Look up the Sun's declination δ for your date (Lecture 1.2) and compute φ. How close is it to the real value?
4. **True north vs compass** (2). Compare your north–south line with a phone compass.
5. **Repeat** (2). Repeat 2–3 weeks later. Did the noon shadow get longer or shorter? Why?

H/L	0.5	0.75	1.0	1.5	2.0	3.0	4.0	6.0	10
h	26.6°	36.9°	45°	56.3°	63.4°	71.6°	76.0°	80.5°	84.3°

Marking: setup and readings (5) · calculations (11) · discussion (4).

Submission checklist

☐ Moon log ($\geq$12 entries) with sketches ☐ graph of % lit vs tithi ☐ shadow-stick table + photo/sketch ☐ answers with working ☐ one paragraph: "What surprised me most"